

Artificial Intelligence CS-202

Topic: Natural Language Processing (NLP)

Habiba Arshad

Department of Computer Science

University of Wah

Week-12

CLO's

This week lecture will cover the following CLO's:

- **CLO1: Discuss** the key concepts in the field of artificial intelligence
- **CLO 2: Implement** artificial intelligence techniques and case studies
- **CLO 3: Model** an automated solution for given problems using artificial intelligence paradigm.

Natural Language processing (NLP)

Natural Language Processing (NLP)

- Natural language processing (NLP) is the ability of a computer program to understand human language as it's spoken and written -- referred to as natural language. It's a component of artificial intelligence (AI).

Key Tasks

- Language translation (e.g., Google Translate).
- Sentiment analysis (e.g., analysing tweets or reviews).
- Chatbots (e.g., customer support bots).

Components of NLP

Components of NLP

Natural Language Processing is not a monolithic, singular approach, but rather, it is composed of several components, each contributing to the overall understanding of language. The main components that NLP strives to understand are:

- Syntax
- Semantics
- Pragmatics
- Discourse.

Components of NLP

Syntax

Definition: Syntax pertains to the arrangement of words and phrases to create well-structured sentences in a language.

Example:

Consider the sentence "The cat sat on the mat."

Syntax involves analyzing the grammatical structure of this sentence, ensuring that it adheres to the grammatical rules of English, such as subject-verb agreement and proper word order.

Components of NLP

Semantics

- **Definition:** Semantics is concerned with understanding the meaning of words and how they create meaning when combined in sentences.

Example:

In the sentence "The panda eats shoots and leaves," semantics helps distinguish whether the panda eats plants (shoots and leaves) or is involved in a violent act (shoots) and then departs (leaves), based on the meaning of the words and the context.

Syntax and Semantics in NLP

- Example:
 - Syntax: "Colorless green ideas sleep furiously" (correct grammar but meaningless semantics).
 - Semantics: "I love programming" (meaningful).



Components of NLP

Pragmatics

Definition: Pragmatics deals with understanding language in various contexts, ensuring that the intended meaning is derived based on the situation, speaker's intent, and shared knowledge.

Example:

If someone says, "Can you pass the salt?"

Pragmatics involves understanding that this is a request rather than a question about one's ability to pass the salt, interpreting the speaker's intent based on the dining context.

Components of NLP

Discourse:

Definition: Discourse focuses on the analysis and interpretation of language beyond the sentence level, considering how sentences relate to each other in texts and conversations.

Example:

In a conversation where one person says, "I'm freezing," and another responds, "I'll close the window," discourse involves understanding the coherence between the two statements, recognizing that the second statement is a response to the implied request in the first.

NLP Applications

NLP Applications and Techniques

- NLP enables machines to understand, interpret, and generate human language.

Core Applications:

- Extracting structured information from text.
- Analysing opinions and emotions.
- Identifying grammar components.

NLP Techniques and Methods

NLP Techniques and Methods

To analyze and understand human language, NLP employs a variety of techniques and methods. Here are some fundamental techniques used in NLP:

- **Tokenization.** This is the process of breaking text into words, phrases, symbols, or other meaningful elements, known as tokens.
- **Parsing.** Parsing involves analyzing the grammatical structure of a sentence to extract meaning.

Text Processing and Preprocessing

Types of Tokenization

- Tokenization can be classified into several types based on how the text is segmented. Here are some types of tokenization:

- **Word Tokenization:**

Word tokenization divides the text into individual words. Many NLP tasks use this approach, in which words are treated as the basic units of meaning.

Example:

- Input: "Tokenization is an important NLP task."
- Output: ["Tokenization", "is", "an", "important", "NLP", "task", "."]

Text Processing and Preprocessing

Types of Tokenization

Sentence Tokenization:

- The text is segmented into sentences during sentence tokenization. This is useful for tasks requiring individual sentence analysis or processing.
- **Example:**
 - Input: "Tokenization is an important NLP task. It helps break down text into smaller units."
 - Output: ["Tokenization is an important NLP task.", "It helps break down text into smaller units."]

Text Processing and Preprocessing

Types of Tokenization

Subword Tokenization:

Subword tokenization entails breaking down words into smaller units, which can be especially useful when dealing with morphologically rich languages or rare words.

Example:

Input: "tokenization"

Output: ["token", "ization"]

Text Processing and Preprocessing

Types of Tokenization

Character Tokenization:

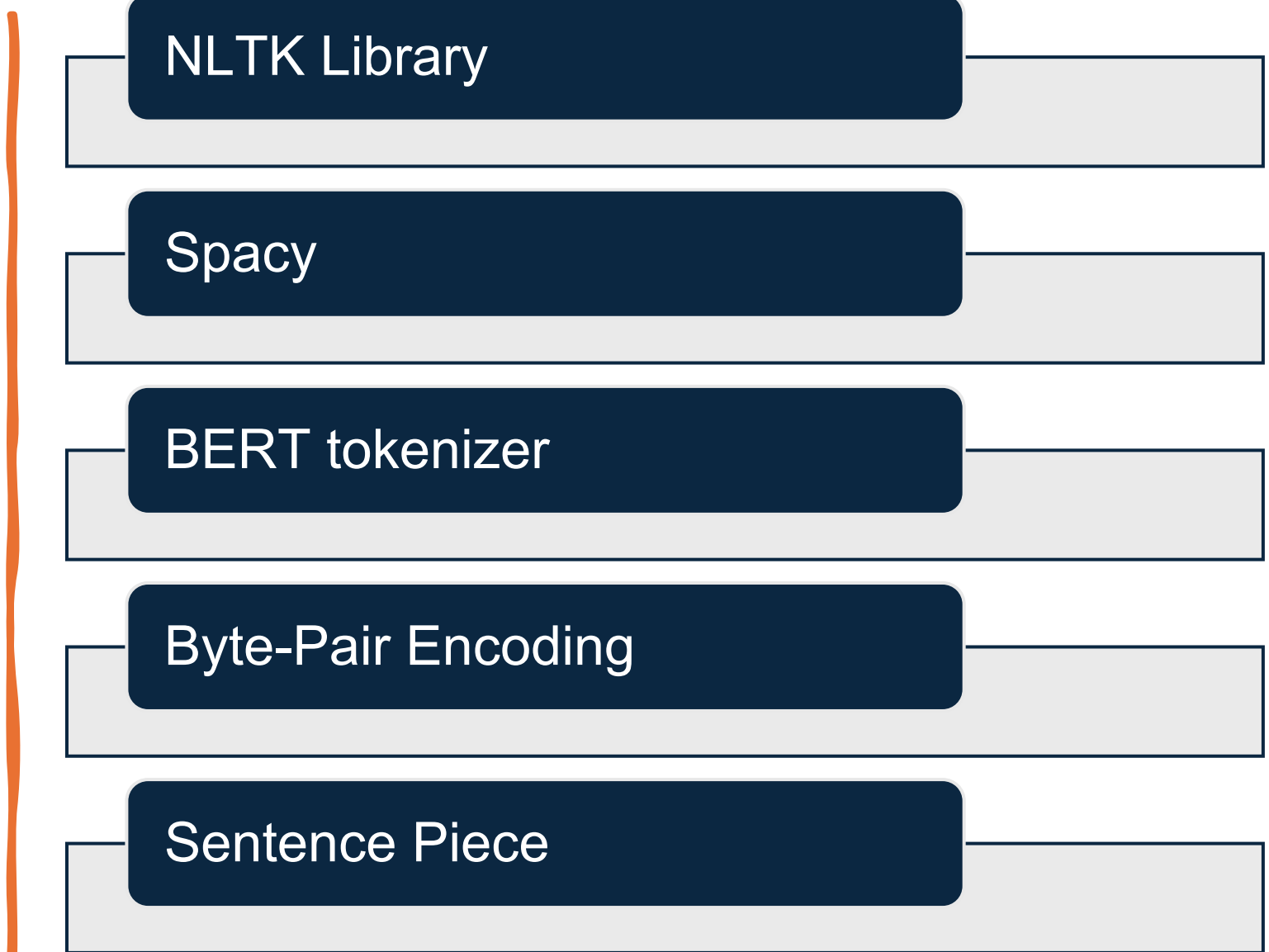
This process divides the text into individual characters. This can be useful for modelling character-level language.

Example:

Input: "Tokenization"

Output: ["T", "o", "k", "e", "n", "i", "z", "a", "t", "i", "o", "n"]

Techniques used for Tokenization



NLTK Library

Spacy

BERT tokenizer

Byte-Pair Encoding

Sentence Piece

Limitations of Tokenization

- Tokenization is unable to capture the meaning of the sentence hence, results in ambiguity.
- In certain languages like Chinese, Japanese, Arabic, lack distinct spaces between words. Hence, there is an absence of clear boundaries that complicates the process of tokenization.
- Text may also include more than one word, for example email address, URLs and special symbols, hence it is difficult to decide how to tokenize such elements.

Stemming and Lemmatization

- Stemming and lemmatization are both text normalization techniques used in Natural Language Processing (NLP) to reduce words to their base or root form. This helps in treating related words (e.g., "run", "running", "runs") as the same in tasks like search, classification, or analysis.

Stemming

- Stemming is the process of chopping off the **end of words** to get to their **root form** (which may not be a real word).
- It uses simple, rule-based heuristics (not always grammatically correct).
- Fast and simple. Useful when perfect accuracy is not necessary (e.g. search engines)
- Can produce non-words
- Less accurate than lemmatization

Stemming

- **Example:**

Original Word	Stemmed Word
running	run
happily	happi
studies	studi
organization	organ

Lemmatization

- This technique reduces words to their base or root form, allowing for the grouping of different forms of the same word.
- It considers context, grammar, and word meaning. Requires a **dictionary or corpus**.
- More accurate and meaningful. Produce real dictionary words
- Slower and complex as it requires more resources.
- Use **lemmatization** when **meaning matters**, like in machine translation, chatbots, or sentiment analysis.

Lemmatization



- Example:

Original Word	Lemma
running	run
studies	study
better	good
was	be



Stop word removal

In natural language processing (NLP), stopwords are frequently filtered out to enhance text analysis and computational efficiency.

Certain words, like "the," "and," and "is," are thought to be ineffective for communicating important information.

The goal of removing stopwords is to speed up text processing by eliminating words that do not add much meaning, even though the set of stopwords can vary.

Types of Stop Words

Common Stopwords: These are the most frequently occurring words in a language and are often removed during text preprocessing. Examples include "the," "is," "in," "for," "where," "when," "to," "at," etc.

Custom Stopwords: Depending on the specific task or domain, additional words may be considered as stopwords. These could be domain-specific terms that don't contribute much to the overall meaning. For example, in a medical context, words like "patient" or "treatment" might be considered as custom stopwords.

Types of Stop Words

Numerical Stopwords: Numbers and numeric characters may be treated as stopwords in certain cases, especially when the analysis is focused on the meaning of the text rather than specific numerical values.

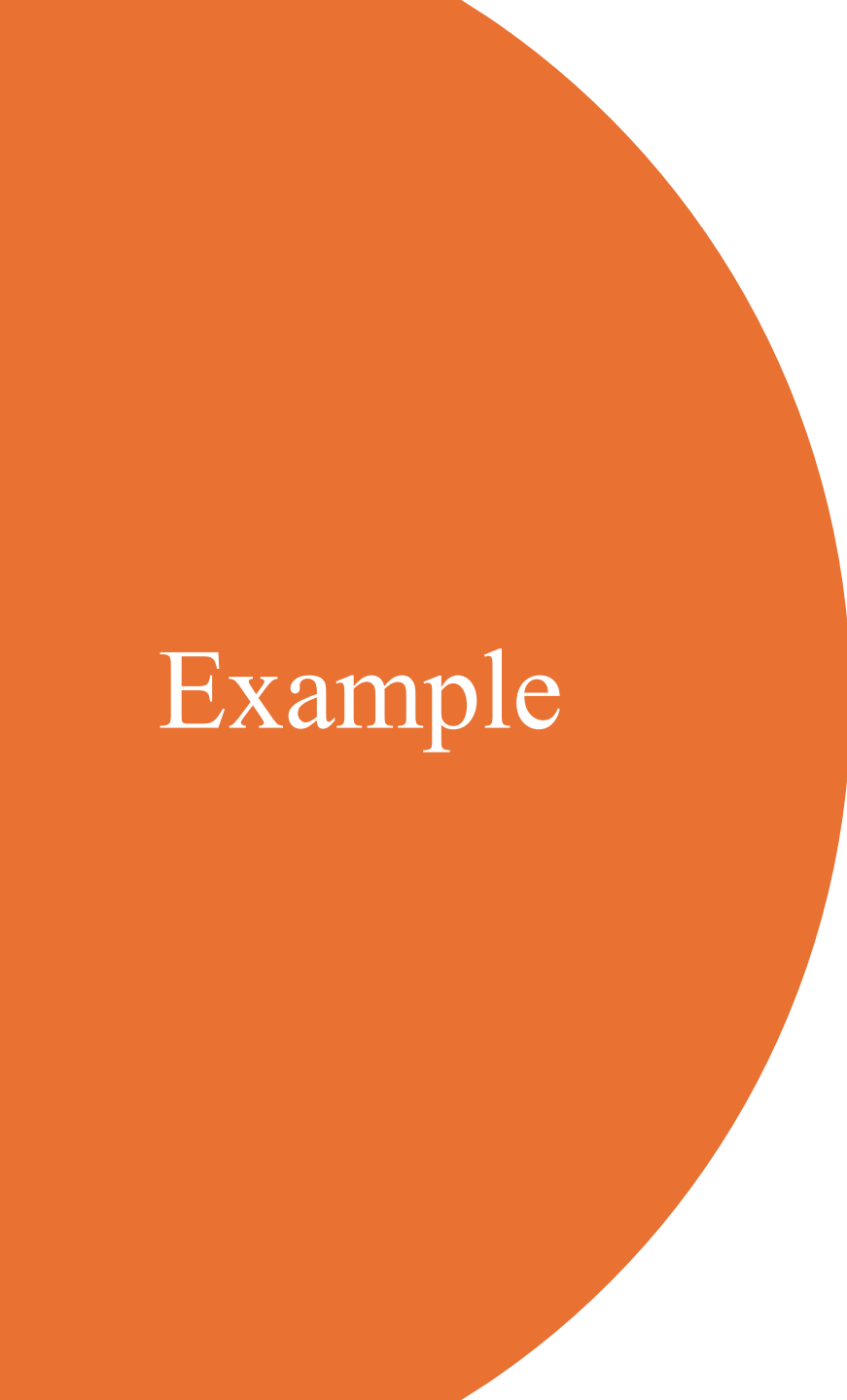
Single-Character Stopwords: Single characters, such as "a," "I," "s," or "x," may be considered stopwords, particularly in cases where they don't convey much meaning on their own.

Contextual Stopwords: Words that are stopwords in one context but meaningful in another may be considered as contextual stopwords. For instance, the word "will" might be a stopword in the context of general language processing but could be important in predicting future events.

Text Normalization

- Text normalization is a crucial preprocessing step in NLP that transforms raw text into a standard, consistent format to improve the performance of downstream tasks like sentiment analysis, translation, or text classification.

Technique	Purpose	Example
Lowercasing	Removes case sensitivity	Hello → hello
Removing punctuation	Cleans text of non-essential symbols	hello! → hello
Removing stop words	Removes frequent, low-meaning words	is, the, an, in → removed
Stemming	Cuts word to root form (may not be valid word)	running → run / runn
Lemmatization	Converts word to base dictionary form	better → good, studies → study
Expanding contractions	Converts short forms to full forms	can't → cannot
Spell correction	Fixes typos and errors	spelng → spelling
Removing numbers/symbols	Cleans up unwanted characters	I have 3 cats → I have cats
Tokenization	Splits text into individual words or sentences	"I love NLP" → ["I", "love", "NLP"]



Example

Text:

"I'm LOVING Natural-Language
Processing"

After Normalization:

i am loving natural language processing



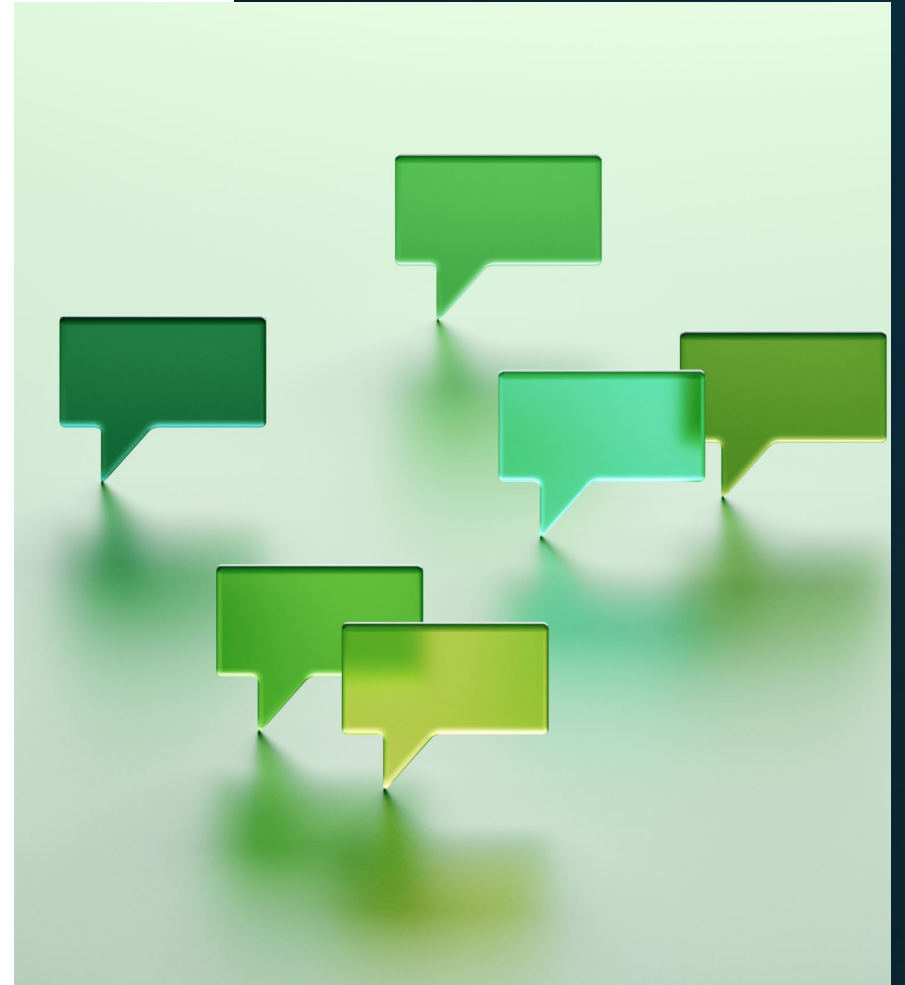
Syntax and Parsing

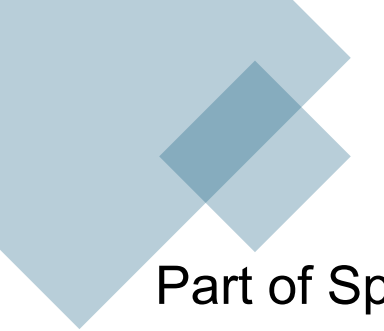
Syntax and Parsing

- **Part-of-Speech (POS) Tagging:** Assigning parts of speech to each word in a sentence (e.g., noun, verb, adjective).
- **Dependency Parsing:** Analyzing the grammatical structure of a sentence to identify relationships between words.
- **Constituency Parsing:** Breaking down a sentence into its constituent parts or phrases (e.g., noun phrases, verb phrases).

POS Tagging

- One of the core tasks in Natural Language Processing (NLP) is Parts of Speech (PoS) tagging, which is giving each word in a text a grammatical category, such as nouns, verbs, adjectives, and adverbs.
- In NLP applications, POS tagging is useful for machine translation, named entity recognition, and information extraction, among other things.
- It also works well for clearing the ambiguity in terms with numerous meanings and revealing a sentence's grammatical structure.





Part of Speech

Simple Definition

Example



Noun

Name of a person, place, thing, or idea

boy, school, book

Pronoun

Used instead of a noun

he, she, it

Verb

Shows action or state

run, eat, is

Adjective

Describes a noun

big, red, happy

Adverb

Describes a verb, adjective, or adverb

quickly, very

Preposition

Shows relation between words
it tells where/when/position

in, on, under

Conjunction

Joins words or sentences

and, but, because

Interjection

Shows sudden feeling

wow!, oh!, hey!

Determiner

If the word tells
which/whose/how many

a, an, the, this, my, some



Common POS Tags

Tag	Part of Speech	Examples
NN	Noun, singular	dog, city, laptop
NNS	Noun, plural	dogs, ideas
VB	Verb, base form	run, go, play
VBD	Verb, past tense	ran, ate
JJ	Adjective	quick, lazy
RB	Adverb	quickly, very
DT	Determiner	the, a, an
IN	Preposition/Subordinating	in, over, because
PRP	Personal pronoun	he, she, it

Example

- Sentence:
"The quick brown fox
jumps over the lazy dog."

Word	POS Tag	Explanation
The	DT	Determiner
quick	JJ	Adjective
brown	JJ	Adjective
fox	NN	Noun (singular)
jumps	VBZ	Verb (3rd person, singular present)
over	IN	Preposition
the	DT	Determiner
lazy	JJ	Adjective
dog	NN	Noun (singular)

Working Example of POS Tagging

- **Example Sentence:** "The cat chased the mouse."

Step 1: Preprocessing the Text

- **Input Text:** "The cat chased the mouse."
- **Tokenization:** Break the sentence into individual words.
Tokens: ["The", "cat", "chased", "the", "mouse"]

Step 2: POS Tagging

- Assign grammatical roles (POS tags) to each word based on their role in the sentence.

Common POS Tags:

- **NN:** Noun (e.g., "cat")
- **VB:** Verb (e.g., "chased")
- **DT:** Determiner (e.g., "The")

Cont....

Step 3: Explanation of Each Tag

- **The:** A determiner (DT), which introduces the noun.
- **cat:** A noun (NN), which is the subject performing the action.
- **chased:** A verb (VBD), indicating the action in the past tense.
- **the:** Another determiner (DT), introducing the next noun.
- **mouse:** A noun (NN), which is the object of the action.

Step 4: How It Works Internally

- **Feature Extraction:**
 - Each word is analysed for:
 - Morphological features (prefixes, suffixes).
 - Context (neighbouring words).
 - Lexical properties (dictionary lookup).

Cont....

Step 5: Model Prediction:

- A trained model (e.g., Hidden Markov Model, Maximum Entropy, or neural network) predicts the POS tag based on these features.

Python Implementation:

```
import nltk
nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')
# Step 1: Input sentence
sentence = "The cat chased the mouse."
# Step 2: Tokenize
tokens = nltk.word_tokenize(sentence)
# Step 3: POS Tagging (Pre-trained model)
pos_tags = nltk.pos_tag(tokens)
# Step 4: Output
print(pos_tags)
```

OUTPUT: [('The', 'DT'), ('cat', 'NN'), ('chased', 'VBD'), ('the', 'DT'), ('mouse', 'NN')]

Semantic Analysis

Named Entity Recognition (NER): Identifying and classifying entities in text into predefined categories , such as :

- Location: countries, cities, landmarks
- Person: names of individuals
- Organization: companies, institutions
- Date/Time: dates, times, durations
- Money/Quantity: currency values, measurements
- Percentages, Events, Products, etc.

Named Entity Recognition (NER) Applications

Use Case

Information Extraction

Question Answering

Chatbots/Assistants

Search Engines

Resume Parsing

Application

Extract key facts from text (e.g., news articles)

Locate specific names, places, or numbers in responses

Understand names, appointments, locations, etc.

Entity-based search (e.g., "Jobs in Google in New York")

Extract names, skills, education, etc.

Example of NER

- "Barack Obama was born in Hawaii in 1961 and served as the President of the United States."

Word/Phrase	Entity Type
Barack Obama	PERSON
Hawaii	LOCATION
1961	DATE
President	TITLE (Optional)
United States	LOCATION

How NER Works?

Text Preprocessing:

- Tokenization, lowercasing, removing stop words.

Feature Extraction:

- Word patterns: Capitalization, prefixes, suffixes.
- Context: Nearby words.

Model Training:

- Supervised learning with labeled data.
- Common models: Conditional Random Fields (CRFs), transformers like BERT.

Entity Identification:

- Classifies words into categories (Person, Organization, Location, etc.).



NER Techniques

Approach	Description
Rule-based	Uses patterns (e.g., capital letters + dictionaries)
Statistical ML	Trained models (e.g., Conditional Random Fields – CRF)
Deep Learning	LSTM, BiLSTM, BiLSTM-CRF models with embeddings
Transformer-based	Uses models like BERT, RoBERTa for high accuracy

Working of NLP

Working of NLP

1. Text Input: Raw human language is input into the system (e.g., a sentence, document, or conversation).

Example: "The movie was amazing and very emotional."

2. Text Preprocessing

Clean and normalize the input text for further analysis.

Step	Description	Example
Tokenization	Splits text into words or sentences	["The", "movie", "was", "amazing"]
Lowercasing	Converts text to lowercase	"the movie was amazing"
Stop Word Removal	Removes common words with little meaning	Removes "the", "was", etc.
Stemming / Lemmatization	Reduces words to root form	"emotional" → "emotion"
POS Tagging	Identifies parts of speech	"movie/NN", "amazing/JJ"

Working of NLP

3. Feature Extraction / Representation:

Convert text into numerical format for machine learning models.

- **Methods:**

- Bag of Words (BoW)

- TF-IDF (Term Frequency – Inverse Document Frequency)

- Word Embeddings (e.g., Word2Vec, GloVe)

- Contextual Embeddings (e.g., BERT)

Example: "amazing" → vector like [0.12, -0.33, 0.85, ...]

Working of NLP

4. Modeling / Learning: Apply machine learning or deep learning to make predictions or extract information.

Depends on task:

- Classification: Spam detection, sentiment analysis
- Sequence Labeling: POS tagging, Named Entity Recognition (NER)
- Generation: Chatbots, summarization
- Translation: Language-to-language conversion

Models used:

Traditional: Naive Bayes, SVM

Deep Learning: Recurrent Neural Network (RNN), Long Short Term Memory (LSTM), Transformers (BERT, GPT)

Working of NLP

5. Output Generation

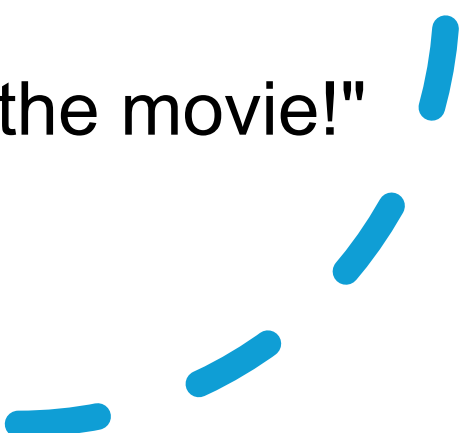
Final output is generated, interpreted, or acted upon.

Could be:

A label: "Positive" (Sentiment Analysis)

A structured format: {"entity": "movie",
"sentiment": "positive"}

A response: "I'm glad you liked the movie!"



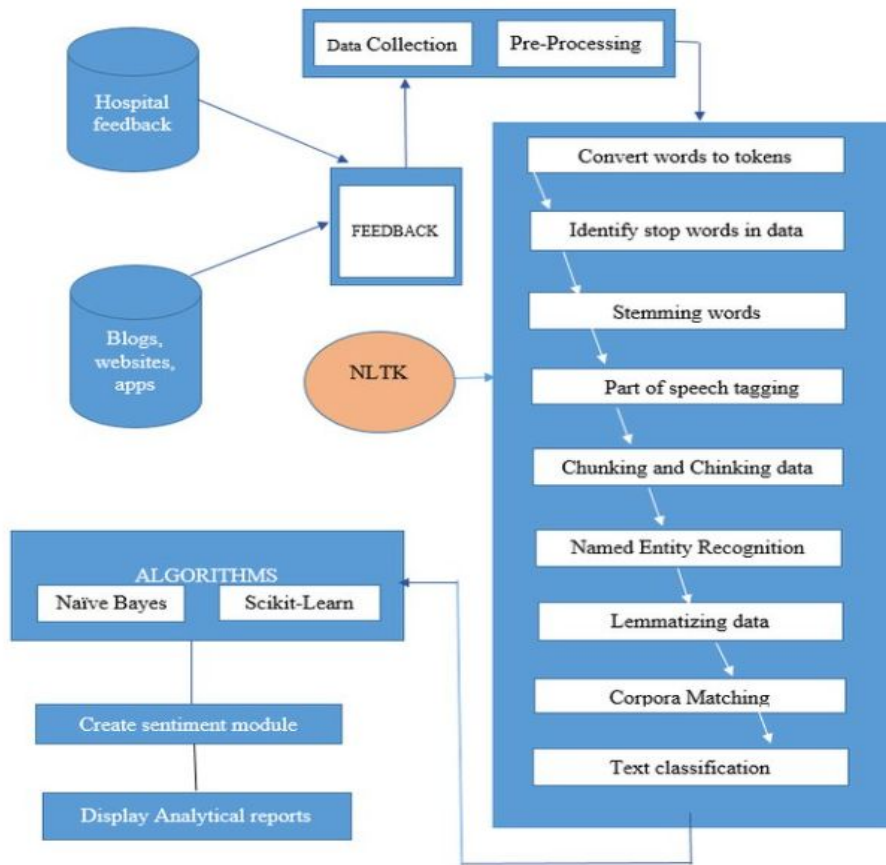
Sentiment Analysis in NLP

- **Sentiment Analysis**, also known as **opinion mining**, is a technique in **Natural Language Processing (NLP)** used to identify and classify the emotions or opinions expressed in a piece of text. Its goal is to determine whether the sentiment of the text is:
- **Positive** (e.g., "I love this movie!"),
- **Negative** (e.g., "I hate this product."),
- **Neutral** (e.g., "The event happened yesterday.").

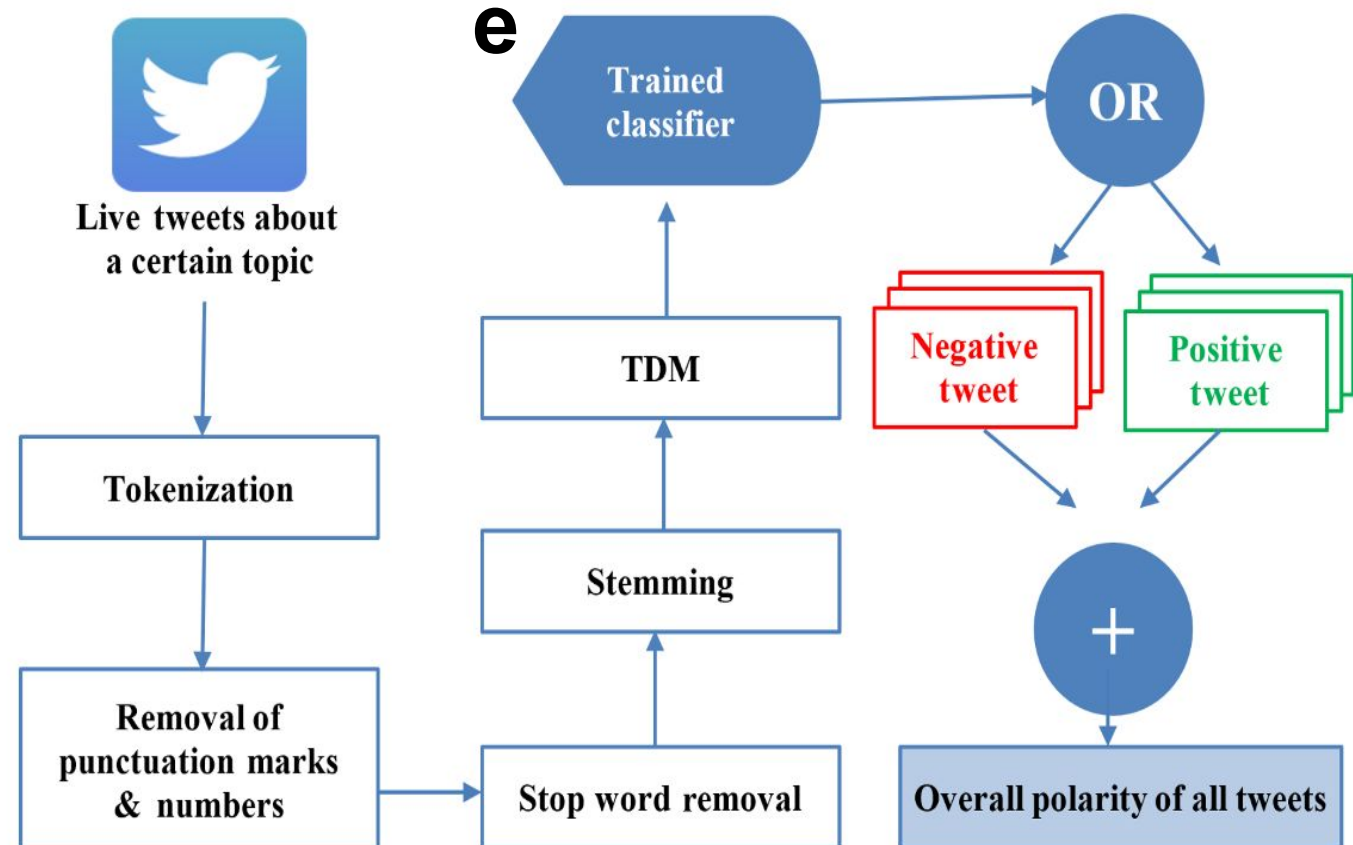


Sentiment Analysis Flow-Chart

General Flow-Chart



Example



Example of NLP & its Explanation

Step 1: Text Preprocessing

- The goal is to prepare the text for analysis by removing unnecessary parts and focusing on key elements.

1. Tokenization:

- Split the sentence into individual words (tokens).
- Tokens: ["The", "movie", "was", "fantastic", "but", "the", "ending", "was", "disappointing"]

2. Stop Word Removal:

- Remove commonly used words like "the", "was", "but", etc., as they don't contribute significantly to the sentiment.
- After removing stop words: ["movie", "fantastic", "ending", "disappointing"]

3. Result of Preprocessing:

- The focus is now on the important words: "movie", "fantastic", "ending", "disappointing".

Explanation Continue

- **Step 2: Sentiment Analysis Model**

- Now, we analyse the sentiment based on the processed text.

1. Word-Level Sentiment Scoring:

1. Each important word is assigned a sentiment score based on the model's trained knowledge:
 1. "fantastic" → Strongly Positive (+2.0)
 2. "disappointing" → Strongly Negative (-1.5)

2. Combining Sentiments:

1. Add up the scores for the words:
 1. Positive score: "fantastic" → +2.0
 2. Negative score: "disappointing" → -1.5
2. Net score = $+2.0 - 1.5 = +0.5$ (positive sentiment dominates).

Explanation Continue

3. Context Analysis:

- The word "but" indicates a contrast. It suggests the negative sentiment ("disappointing") applies only to the ending, not the entire movie.
- The positive word "fantastic" strongly impacts the overall sentiment because it describes the movie as a whole.

4. Classification:

- The sentiment analysis model uses a threshold to classify the overall sentiment:
 - If the score is above 0, it's **positive**.
 - If the score is below 0, it's **negative**.
- Since the net score is +0.5, the sentence is classified as **positive**.



Why is it Positive?

- The positive sentiment ("fantastic") about the movie as a whole outweighs the negative sentiment ("disappointing") about the ending.
- Words like "fantastic" carry a stronger weight in sentiment analysis, especially when they refer to the broader context of the sentence (the movie overall).

Key Applications of NLP



Machine Translation (e.g., Google Translate).



Virtual Assistants (e.g., Alexa, Siri).



Sentiment Analysis (e.g., product reviews).



Information Retrieval (e.g., search engines).



Text Summarization (e.g., summarizing articles).

Challenges in NLP

Ambiguity

Example: "He saw the man with the telescope" (Who has the telescope?).

Context Understanding

Example: "Bank" (Riverbank vs. Financial bank).

Multilingual Texts

Processing multiple languages effectively.

Noisy Data

Slang, typos, and informal text.
